TIME APPLIANCES (TAP)



Multidomain Time Synchronization in the Modern Datacenter: From Network to Workload



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Accuracy and Availability at Datacenter Scale



ACCURACY

Improve time synchronization accuracy for datacenter workloads

- Databases – reduce commit-wait period yielding higher throughput
- AI workloads – reduce cluster idle time, resulting in faster training and lower power usage



AVAILABILITY

Maximize time availability (e.g., 99.9999%)

- Leverage hot-standby technique to improve failover performance

Improved Accuracy using Multiple Timescales

Issue: Time accuracy degrades with each hop between the Grandmaster (GM) and end-station

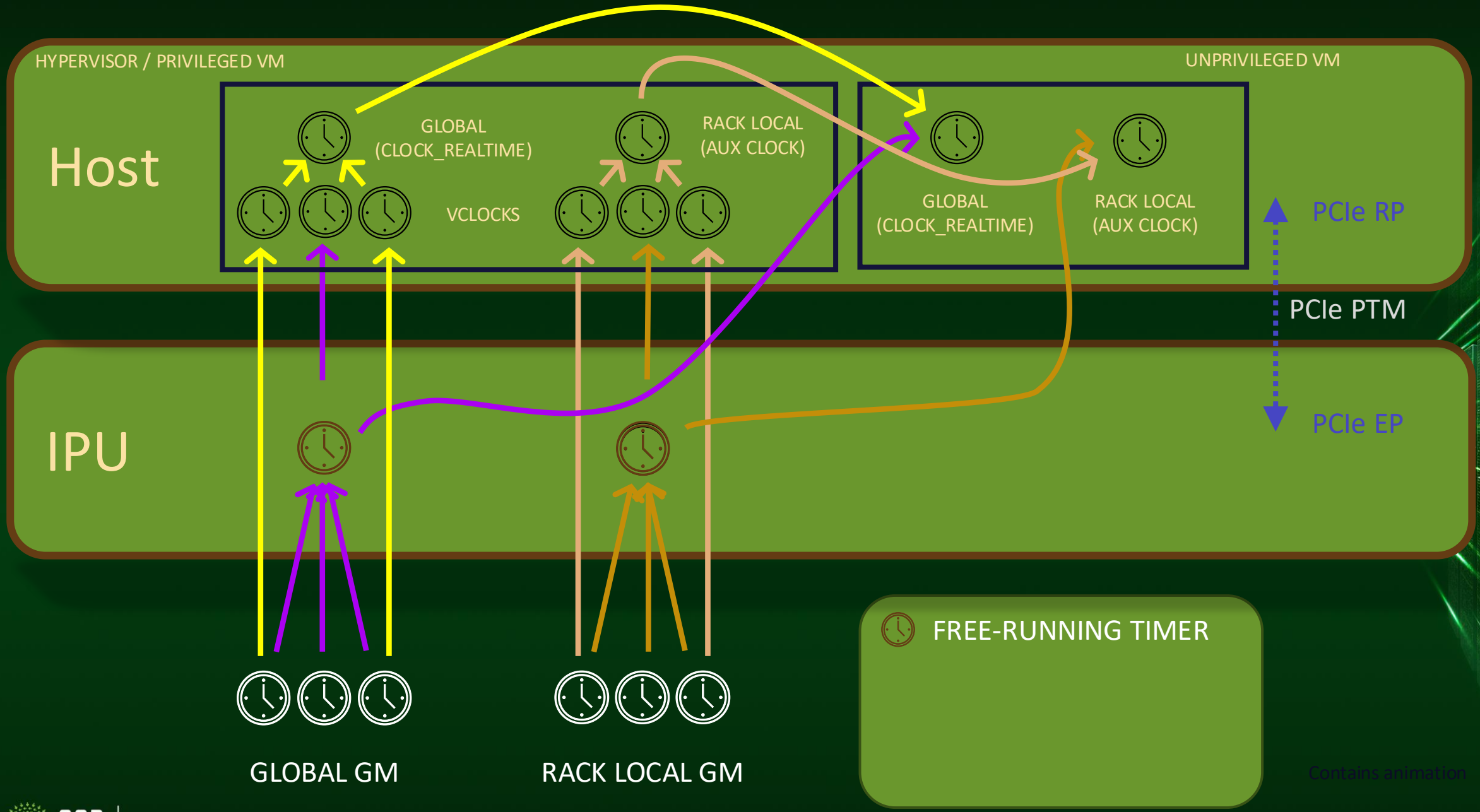
Solution: Introduce multiple timescales with different hop counts

Rack Local timescale

- For intra-rack task synchronization, application event logging
- Per-rack Timescale: Improved accuracy with a maximum of two hops from GM to end station
- Potential for better than 50 nanosecond synchronization to the software visible OS clock
- Not necessarily traceable to TAI
- Mapped as an auxiliary clock (auxclock) in Linux
  - Accessed using POSIX clock API

Global timescale

- For inter-rack task synchronization and global event logging
- Larger maximum hop count limit
- Used for standard OS timekeeping
- Traceable to GNSS/UTC
- Typically mapped to CLOCK_REALTIME in Linux


HYPERVISOR / PRIVILEGED VM
UNPRIVILEGED VM
Host
GLOBAL
(CLOCK_REALTIME)
RACK LOCAL
(AUX CLOCK)
VCLOCKS
GLOBAL
(CLOCK_REALTIME)
RACK LOCAL
(AUX CLOCK)
PCIe RP
PCIe PTM
PCIe EP
IPU
GLOBAL GM
RACK LOCAL GM
FREE-RUNNING TIMER
Contains animation

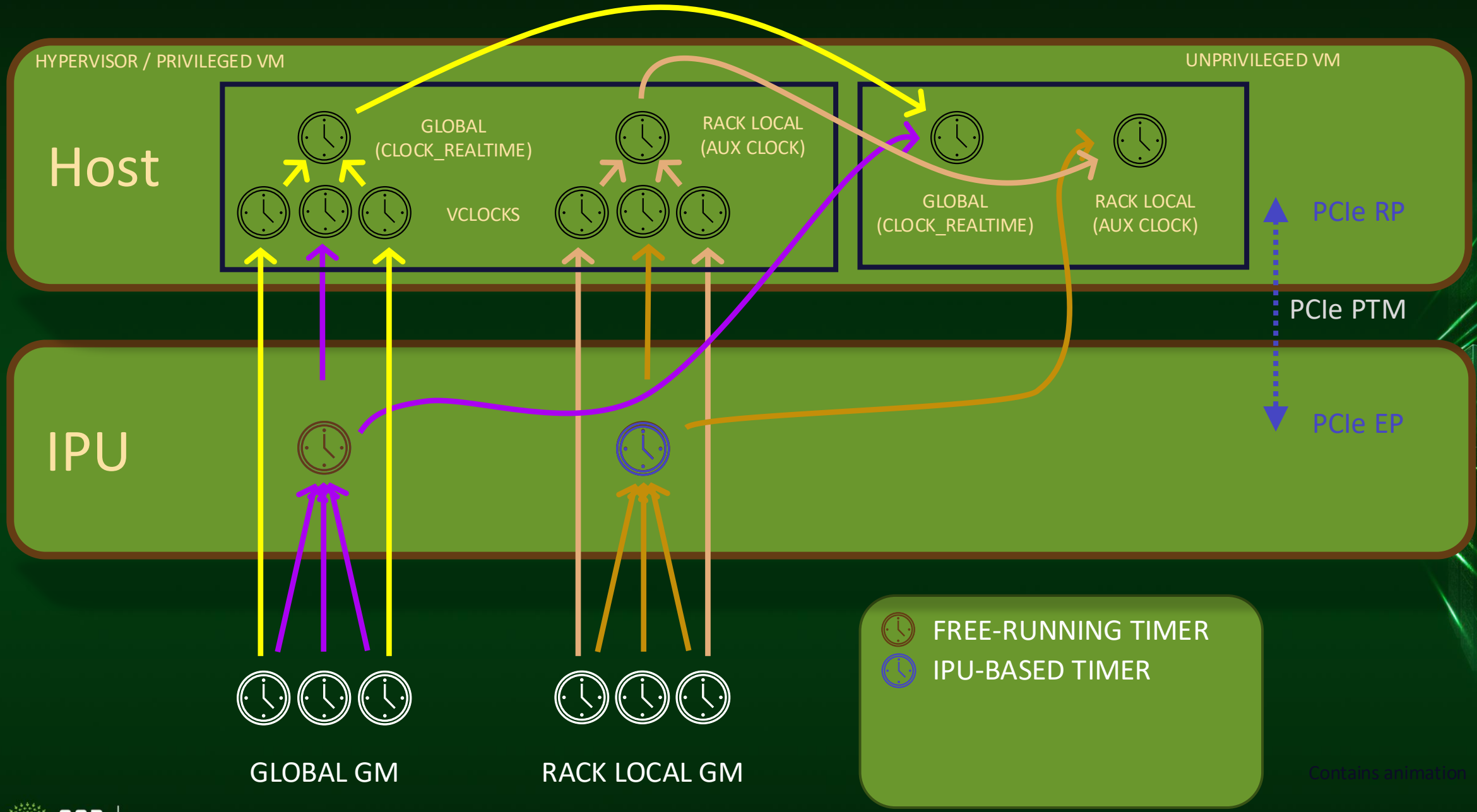
Time Virtualization

Issue: VMs do not have direct access to PTP time

Solution: Share time within the platform using TSC as a common reference

OS-Level VM Clock Sharing

- Uses TSC as a common clock reference to share clocks across VM instances
- Already limited support in the Linux kernel
  - PTP KVM
  - Requires generalization to other VM architectures
- Currently, requires hypervisor support and, in general, requires a VM exit to discipline PTP time


HYPERVISOR / PRIVILEGED VM
UNPRIVILEGED VM
Host
GLOBAL
(CLOCK_REALTIME)
RACK LOCAL
(AUX CLOCK)
VCLOCKS
GLOBAL
(CLOCK_REALTIME)
RACK LOCAL
(AUX CLOCK)
PCIe RP
PCIe PTM
PCIe EP
IPU
GLOBAL GM
RACK LOCAL GM
FREE-RUNNING TIMER
IPU-BASED TIMER
Contains animation

Hot Standby

Issue: BTCA (BMCA) takes several seconds switch over to the new GM when the current GM fails,

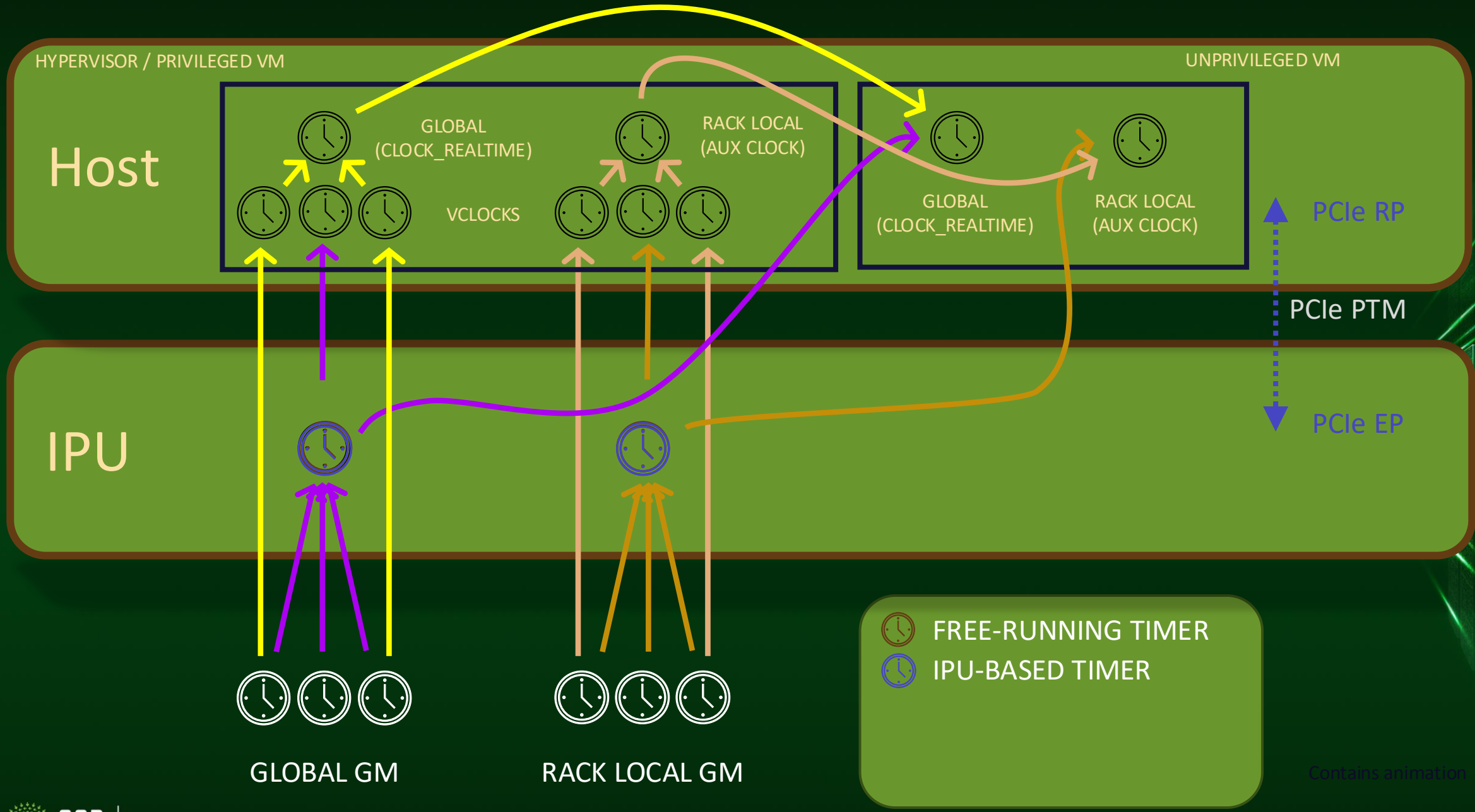
Solution: Synchronize to multiple GMs simultaneously, when a GM fails switch to a backup GM

Host Endpoint

- The host simultaneous syncs with multiple GMs
- Each backup GM is replicated using a vclock
- Host software detects when a source has disappeared and immediately switches over to the backup
- Backup GMs must remain closely synchronized with the primary GM, otherwise large time skews between end-stations may occur
- Clock Manager (part of libptpmgmt) provides clock monitor function to implement hot standby

IPU Endpoint

- The IPU connects to multiple GMs simultaneously
- IPU firmware detects when a source has disappeared and switches to a backup time source
- To the host, the composite clock appears as a single PHC timesource
- Offloads redundancy implementation from the host
- In virtualized environments, there is no need for PTP support in a service VM



Contains animation

Call to Action and Future Work

Complete support for auxclocks in Linux kernel, Clock Manager

- Clock Manager: https://github.com/erezgeva/libptpmgmt/tree/master/clkmgr
- Linux auxclock: https://lore.kernel.org/all/20250519082042.742926976@linutronix.de/

Support for PTM in NICs and host platform

Propose improvements to the OCP DataCenter (OCP-DC)

PTP profile supporting multidomain and hot-standby redundancy in alignment with 802.AS-2020 and 802.1ASdm

Improved redundancy

- Byzantine fault-tolerance – resilience to *attack* as well as failure




Thank
You!
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Improved Synchronization Accuracy using PTM

Issue: Precision is lost replicating IPU device clock using software only methods

Solution: Use the PTM protocol to improve synchronization on the PCIe connection between host and NIC

Background

- PTM messages are used by the downstream device to measure the link delay for each PCIe link using RX/TX timestamps from the local and upstream device clocks
- PTM Response message sent from the upstream device contains PTM root time – similar to a sync message
- Downstream device compute its offset from the upstream device using:
  1. PTM root time in the response message
  2. link delay
  3. local arrival timestamp

Use of PTM on the IPU

- Use PTM to create a shared clock reference between the host and IPU
- This reference is used as a basis for the Global and Rack-local Clocks
- VMs also use the shared clock reference for direct access to the Global and Rack-local clocks